

CODEALPHA INTERNSHIP PROGRAM

IoT Internship – Task 4: Mini Project (IoT Case Study)

IoT Security Challenges:

Risks, Real-World Attacks, and

Future Protection Strategies

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1. EXECUTIVE SUMMARY

This report documents the design, implementation, and validation of a **Smart Home Temperature Monitoring and Automatic Cooling System**, developed as Task 2 of the CodeAlpha IoT Internship Program. The project simulates a real-world Internet of Things (IoT) application using the Arduino Uno microcontroller platform within the Tinkercad Circuits virtual environment. The system continuously monitors ambient room temperature and generates multi-modal alerts—visual and auditory—when readings exceed a predefined safety threshold of 30°C. This project demonstrates fundamental IoT principles including sensor data acquisition, analog-to-digital conversion, threshold-based decision making, and real-time actuation across multiple output devices.

2. PROJECT OBJECTIVES

The primary objectives of this project are:

1. **Continuous Temperature Sensing:** Read ambient temperature values from a TMP36 analog sensor every one second.
 2. **Real-Time Visual Feedback:** Display current temperature readings and system status on a 16×2 LCD screen.
 3. **Automated Alert System:** Trigger visual (LED) and audio (buzzer) warnings when temperature crosses the 30°C threshold.
 4. **Educational Demonstration:** Showcase core IoT architectural concepts—Sensing, Processing, and Actuation—in a simulated environment.
 5. **System Reliability:** Ensure accurate sensor readings, stable component operation, and predictable behavior across the operating range.
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3. SYSTEM ARCHITECTURE

The project adopts a classical **three-layer IoT architecture** that mirrors real-world smart home deployments:

3.1 Sensing Layer

The **TMP36 analog temperature sensor** forms the sensing layer. This precision sensor outputs a linearly proportional voltage relative to ambient temperature, with a scale factor

of 10 mV per °C and an offset voltage of 500 mV at 0°C. Its operating range of –40°C to +125°C comfortably covers residential monitoring requirements.

3.2 Processing Layer

The **Arduino Uno R3**, based on the ATmega328P microcontroller, serves as the processing layer. Its built-in 10-bit Analog-to-Digital Converter (ADC) samples the TMP36 output voltage, converting continuous analog signals into discrete digital values (0–1023). The microcontroller then applies the TMP36 conversion formula and executes conditional logic to determine appropriate system responses.

3.3 Output/Actuation Layer

The output layer delivers user notifications through four devices:

- **16×2 Liquid Crystal Display (LCD):** Textual temperature and status information
- **Green LED:** Visual indicator for normal operating conditions
- **Red LED:** Visual indicator for high-temperature alerts
- **Piezoelectric Buzzer:** Audible 1000 Hz alarm during alert conditions

4. HARDWARE COMPONENTS

#	Component	Quantity	Function
1	Arduino Uno R3	1	Central microcontroller
2	TMP36 Temperature Sensor	1	Ambient temperature sensing
3	16×2 LCD Display	1	Visual information display
4	Green LED	1	Normal-state indicator
5	Red LED	1	Alert-state indicator
6	Piezo Buzzer	1	Audio alert generator
7	220Ω Resistors	3	Current limiting (LEDs + LCD backlight)
8	10kΩ Potentiometer	1	LCD contrast adjustment
9	Breadboard	1	Component interconnection platform

#	Component	Quantity	Function
10	Jumper Wires	~25	Electrical connections

5. CIRCUIT DESIGN AND WIRING

5.1 Pin Configuration Summary

Arduino Pin	Connected Component	Signal Type
A0	TMP36 Vout	Analog Input
Pin 2	LCD D7	Digital Output
Pin 3	LCD D6	Digital Output
Pin 4	LCD D5	Digital Output
Pin 5	LCD D4	Digital Output
Pin 8	Green LED (via 220Ω)	Digital Output
Pin 9	Red LED (via 220Ω)	Digital Output
Pin 10	Piezo Buzzer	PWM Output
Pin 11	LCD Enable	Digital Output
Pin 12	LCD Register Select	Digital Output
5V	Power distribution	Power
GND	Ground distribution	Ground

5.2 Key Wiring Details

The TMP36 sensor connects with its left pin to 5V, middle pin to Arduino A0, and right pin to GND. Both LEDs use 220Ω current-limiting resistors in series between the Arduino digital pin and the LED anode. The LCD operates in 4-bit mode, minimizing pin usage from eight to four data lines. A 10kΩ potentiometer, wired as a voltage divider between 5V and GND, feeds LCD Pin 3 (V0) to control display contrast. The LCD backlight requires its own 220Ω resistor to prevent excessive current draw.

6. SOFTWARE IMPLEMENTATION

6.1 Code Structure

The Arduino program follows a modular design comprising five key functions:

1. `setup()` — Initializes serial communication, configures pin modes, sets initial output states, initializes the LCD, registers the custom degree symbol, and displays the startup message.
2. `loop()` — The main execution cycle that calls all sub-functions and enforces the 1-second update interval.
3. `readTemperature()` — Reads analog data from A0, converts it to voltage, then applies the TMP36 formula to obtain temperature in Celsius.
4. `updateLCDDisplay()` — Refreshes temperature value on row 1 and status message on row 2.
5. `controlOutputs()` — Executes threshold logic to control LEDs and buzzer.
6. `printToSerial()` — Outputs diagnostic data to the Serial Monitor for debugging.

6.2 Temperature Conversion Mathematics

Two sequential calculations transform the raw ADC reading into temperature:

text

Step 1: ADC → Voltage

$$\text{Voltage} = (\text{ADC_reading} / 1023) \times 5.0$$

Step 2: Voltage → Temperature

$$\text{Temperature}(\text{°C}) = (\text{Voltage} - 0.5) / 0.01$$

Worked Example: For a raw ADC reading of 154:

- $\text{Voltage} = (154 / 1023) \times 5.0 = 0.753 \text{ V}$
- $\text{Temperature} = (0.753 - 0.5) / 0.01 = 25.3\text{°C}$

6.3 Threshold Logic

C++

```
if (currentTemperature >= 30.0) {  
    // Alert Mode  
    digitalWrite(GREEN_LED_PIN, LOW);  
    digitalWrite(RED_LED_PIN, HIGH);  
    tone(BUZZER_PIN, 1000);  
} else {  
    // Normal Mode  
    digitalWrite(GREEN_LED_PIN, HIGH);  
    digitalWrite(RED_LED_PIN, LOW);  
    noTone(BUZZER_PIN);  
}
```

7. FUNCTIONAL OPERATION

7.1 Normal Mode (Temperature < 30°C)

- LCD Row 1: Temp: XX.X°C
- LCD Row 2: Status: Normal
- Green LED: **ON** ✓
- Red LED: **OFF**
- Buzzer: **Silent**

7.2 Alert Mode (Temperature ≥ 30°C)

- LCD Row 1: Temp: XX.X°C
- LCD Row 2: Status: HIGH TEMP!
- Green LED: **OFF**
- Red LED: **ON** ✓
- Buzzer: **1000 Hz tone active**

7.3 System Timing

The main loop executes at a 1-second interval, striking an optimal balance between responsiveness and processing efficiency. This rate is suitable for monitoring gradual environmental temperature changes typical in residential settings.

8. TESTING AND VALIDATION

Comprehensive testing was performed across multiple scenarios:

Test Case	Set Temperature	Expected Behavior	Result
Startup Test	N/A	Welcome message displayed 2s	 Pass
Low Temp	15°C	Green LED, "Normal"	 Pass
Normal Temp	25°C	Green LED, "Normal"	 Pass
Threshold Edge	29.9°C	Green LED, "Normal"	 Pass
Threshold Trigger	30.0°C	Red LED, Buzzer, "HIGH TEMP!"	 Pass
High Temp	45°C	Red LED, Buzzer, "HIGH TEMP!"	 Pass
Extreme Temp	100°C	Red LED, Buzzer, "HIGH TEMP!"	 Pass
Serial Output	Any	Correct data every 1s	 Pass

Sample Serial Monitor Output:

text

Temp: 24.8 C | Voltage: 0.748V | Status: Normal

Temp: 30.2 C | Voltage: 0.802V | Status: HIGH TEMPERATURE ALERT!

Temp: 35.5 C | Voltage: 0.855V | Status: HIGH TEMPERATURE ALERT!

All tests confirmed the system behaves precisely according to specifications, with accurate temperature conversions and correct actuator responses.

9. CHALLENGES AND SOLUTIONS

Challenge 1: Incorrect TMP36 Readings

Early testing produced impossible values (261°C). Investigation revealed that the signal wire was inadvertently connected to a shared column with the potentiometer, causing voltage bleed-through. **Solution:** The signal wire was carefully re-routed directly from the TMP36 middle pin to Arduino A0, resulting in accurate readings.

Challenge 2: LCD Blank Display

Initially, the LCD showed blank screens despite being powered. **Solution:** Adjusting the 10kΩ potentiometer to approximately 25% of its rotation range provided optimal contrast, revealing clear text.

Challenge 3: Excessive Backlight Current

Tinkercad flagged a critical warning about 23.7 kA current through the LCD backlight due to a direct 5V connection. **Solution:** A 220Ω current-limiting resistor was inserted between LCD Pin 15 and the 5V rail, safely limiting current to acceptable levels.

10. WHY THIS IS AN IoT SIMULATION

Although developed in a virtual environment, this project qualifies as a legitimate IoT simulation for several reasons:

1. **Identical Architecture:** The Sense → Process → Actuate flow mirrors real IoT deployments.
 2. **Accurate Sensor Modeling:** Tinkercad simulates TMP36 using its actual datasheet characteristics.
 3. **Portable Code:** The Arduino sketch runs unchanged on physical hardware.
 4. **Complete IoT Components:** All architectural layers (sensing, computing, actuation, monitoring) are functional.
 5. **Industry Practice:** Digital simulation is widely used by professionals for pre-deployment validation, reducing cost and risk.
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11. FUTURE IMPROVEMENTS

Several enhancements could extend the project into a production-grade system:

- **WiFi Connectivity:** Replace Arduino Uno with ESP8266/ESP32 to enable cloud data upload via MQTT or HTTP.

- **Data Logging:** Add SD card storage for historical temperature analysis.
 - **Mobile Integration:** Push notifications via Blynk, Firebase, or Twilio SMS.
 - **Automatic Cooling:** Integrate a relay module to control an actual fan or air conditioner.
 - **Multi-Zone Monitoring:** Deploy multiple TMP36 sensors with a multiplexer for room-by-room monitoring.
 - **Predictive Analytics:** Implement TinyML models to predict temperature trends and pre-activate cooling.
 - **Enhanced Display:** Upgrade to a 128×64 OLED with real-time graphical temperature trends.
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12. LEARNING OUTCOMES

Through this project, the following key skills and knowledge were acquired:

- **Analog Sensor Interfacing:** Practical understanding of ADC operations and voltage-to-physical-quantity conversions.
 - **LCD Programming:** Mastery of the LiquidCrystal library, 4-bit communication mode, and custom character creation.
 - **Multi-Output Coordination:** Simultaneous management of multiple output devices based on sensor input.
 - **Circuit Design Principles:** Proper use of current-limiting resistors, breadboard organization, and power distribution.
 - **Debugging Techniques:** Systematic troubleshooting using Serial Monitor and simulated instrumentation.
 - **IoT Architectural Thinking:** Application of layered IoT design patterns in a working system.
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13. CONCLUSION

The Smart Home Temperature Monitoring and Automatic Cooling System successfully fulfills all functional requirements outlined by CodeAlpha Task 2. The project demonstrates comprehensive integration of an analog sensor, digital microcontroller, and multiple

actuators to create a responsive, real-time monitoring solution. Every test case passed validation, confirming the system's reliability and accuracy.

Beyond meeting technical specifications, this project reinforced foundational IoT principles that translate directly to real-world applications in smart homes, industrial monitoring, agriculture, and healthcare. The Tinkercad simulation environment proved invaluable for rapid prototyping, providing risk-free experimentation and instant validation. The modular code architecture and organized circuit design also ensure the project can be readily extended with additional features such as network connectivity, cloud integration, and machine learning capabilities.

This exercise not only strengthened technical proficiency in embedded systems and Arduino programming but also cultivated critical debugging skills, systematic problem-solving approaches, and appreciation for professional engineering documentation practices—all essential competencies for a career in IoT development.

14. REFERENCES

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5. HD44780 LCD Controller Datasheet — Hitachi